

Experiment – 04

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Subject: Advanced Database
Management System

Question 1:

- (i) You are given a table tbl_happiness that contains the ranking of happiness for different countries. Your task is to retrieve all the records from the table while ensuring that "India" and "Sri Lanka" appear at the top, followed by the rest of the countries in their original ranking order.
- (ii) Make all the rankings in descending order except India and Sri Lanka.

S.No	Rankings	Country
1	1	Finland
2	2	Denmark
3	3	Iceland
4	4	Israel
5	5	Netherlands
6	6	Sweden
7	7	Norway
8	126	India
9	128	Sri Lanka

Output:

Rankings	Country
126	India
128	Sri Lanka
1	Finland
2	Denmark
3	Iceland
4	Israel

Rankings	Country
126	India
128	Sri Lanka
7	Norway
6	Sweden
5	Netherlands
4	Israel
3	Iceland
2	Denmark
1	Finland

Solution:

```
SELECT
  RANKINGS,
  COUNTRY
FROM TBL_HAPPINESS
ORDER BY
  CASE
    WHEN COUNTRY = 'India' THEN 1
    WHEN COUNTRY = 'Sri Lanka' THEN 2
    ELSE 3
  END,
  RANKINGS;
```

```
SELECT
  RANKINGS,
  COUNTRY
FROM TBL_HAPPINESS
ORDER BY
  CASE
    WHEN COUNTRY = 'India' THEN 1
```

```

    WHEN COUNTRY = 'Sri Lanka' THEN 2
    ELSE 3
END,
CASE
    WHEN COUNTRY NOT IN ('India', 'Sri Lanka')
    THEN RANKINGS
END DESC;

```

Question 2:

A company maintains employee information in a single table where each employee may report to another employee acting as their manager. Prepare a report that displays the details of every employee who supervises at least one other employee. The report should include the manager's employee ID, name, total number of direct reports, and the average age of those reporting employees. The average age should be rounded to the nearest whole number and the final result should be sorted by employee ID.

employee_id	name	reports_to	age
9	Hercy	NULL	43
6	Alice	9	41
4	Bob	9	36
2	Winston	NULL	37

Output:

employee_id	name	reports_count	average_age
9	Hercy	2	39

Solution:

```

SELECT
    E1.EMPLOYEE_ID,
    E1.NAME,
    COUNT(E2.EMPLOYEE_ID) AS REPORTS_COUNT,
    (
        SELECT ROUND(AVG(AGE), 0)
        FROM EMPLOYEES
        WHERE REPORTS_TO = E1.EMPLOYEE_ID
    ) AS AVERAGE_AGE
FROM EMPLOYEES AS E1
JOIN EMPLOYEES AS E2
    ON E1.EMPLOYEE_ID = E2.REPORTS_TO
GROUP BY
    E1.EMPLOYEE_ID,
    E1.NAME
ORDER BY
    E1.EMPLOYEE_ID;

```